Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Teacher and cover plan

1. Lesson outcome SCI-LAUNCH-A1-W08-L2 · Science · Autumn 1 · Week 8 I can use continuous-sampling evidence to investigate how pH affects amylase activity.

2. Arrival 1–4 Four questions; see STAFF ANSWERS · Arrival and the matching tier sheet.

3. Starter Which setup protects the first observations?

4. I Do I do · follow the sampled drops; I do · record what was observed. In your book: Watch the model. No writing yet.

5. We Do We do · method-decision checkpoint; We do · repair an evidence claim. In your book: Record one evidence claim and explain its limit.

6. You Do Complete the chosen route using the shared evidence. In your book: Improve one explanation; mark the change // and read it back.

7. Support routes Independent → Supported → Choice; Aspire separately. Staff mark I, S or C beside this task.

8. Adaptations / normal ways of working Use planned adaptations in TA brief. Private response, seated or adult-directed access; agree a pause and return step.

9. TA focus No cold calling; nobody is asked to read difficult text aloud. Keep the first response. Use the ladder, then fade support.

10. Exit Two checks; use the matching tier. Staff guidance: STAFF ANSWERS · Exit.

11. After teaching Misconception: _____ Response: _____ Next step: _____

12. Preparation / evidence / cover essentials Use supplied models, cards, data and book. Read TA brief: Preparation and practical safety before teaching. Staff answers: Arrival, Exit, shared tasks and mark scheme. Use the supplied models, cards, data and booklet. This lesson plans only. During the later practical, handle reagents only with the centre-approved method, prepared equipment, local risk assessment and trained supervision. The supplied-observation route needs no reagents and teaches the same interpretation; it does not certify physical practical completion. Do not infer concentrations, heating arrangements or disposal from lesson cards. Further safety and preparation: continued on page 2.

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Teacher and cover plan · timeline and support

Prompt ladder for adults 1. Point to the relevant place / instruction. 2. Ask one short question: "What is it asking you to find?" 3. Rephrase the instruction in simpler language. 4. Provide a scaffold or limited choice. 5. Model the first step only, then return the task to the pupil.
00–03 · Arrival task · 3 min TA: Help locate and begin. Do not answer the questions.
03–07 · Starter · ready before mixing · 4 min TA: Help access the question. Do not provide the response.
07–11 · I do · follow the sampled drops · 4 min TA: Support attention. Do not give a parallel explanation.
11–15 · We do · method-decision checkpoint · 4 min TA: Prompt contribution and recording. Do not complete the guided response.
15–19 · I do · record what was observed · 4 min TA: Support attention. Do not give a parallel explanation.
19–24 · We do · repair an evidence claim · 5 min TA: Prompt contribution and recording. Do not complete the guided response.
24–34 · You do · 10 min · KEEP TA: Use the ladder; record help and return to independence. Do not give the content answer.
34–37 · Review and improve · 3 min · KEEP TA: Receive the pupil-owned change. Do not change the pupil's meaning.
37–40 · Exit ticket · 3 min · KEEP TA: Ensure an attempt. Do not change an answer.
KEEP / shorten Keep independent evidence, review and exit. Take one shared explanation if short of time.
Evidence adult Teacher or TA receives the explanation; record actual support. Keep raw observations distinct from illustrative data. Use the lesson code on the evidence app.
Preparation and safety · continued Choose a route at each task using what the pupil shows. Switching adds no task, record or label. Risk-note controls cross-checked against publicly accessible [SYS] [QUAL] Enzymes and [QUAL] Iodine on 26 September 2026. Teacher/technician must authorise the local method, concentrations, quantities, PPE and disposal using current member guidance. The full member-only [QUAL] method and [SYS] were not accessible for this check. Use minimum enzyme and iodine concentrations and quantities. Prevent enzyme powder aerosol; do not spray solutions. Follow eye-protection and solvent-specific iodine controls. Illustrative or supplied data is reasoning practice only and is never logged as practical completion. Log only an observed, authorised physical contribution; the teacher decides whether it meets the relevant course requirement.